

* ADC programming with STM 32 ARM:

↳ The STM32F4xx ARM chip has three on-chip ADC modules.
ADC1 (Master), ADC2 (slave) and ADC3 (slave)

↳ It can support upto 16 external analog input channels.

↳ The ADC has 12-bit resolution

↳ Major Registers for ADC programming:

→ ADC_SR (Status Register)

→ ADC_CR (Control Register)

→ ADC_SMPR (Sample Time Register)

→ ADC_DR (Data Register): After conversion, the result is placed in the Data Register.

↳ The ADC needs a clock to drive the conversion

↳ The ADC clock is enabled by bits of RCC_APB2ENR (RCC_APB2 peripheral clock enable Register) in the

Reset and control (RCC) section of the CPU

↳ The Conversion clock is derived from the SysCLK.

* STM32 ARM on-chip DAC :-

- ↳ STM32F4xx ARM chips have on-chip DAC
- ↳ Having one on-chip DAC
- ↳ PA4 → is the analog o/p pin
- ↳ It has 12-bit Resolution but it can also be configured as 8-bit DAC too. i.e. digital data loaded into DAC register can be 8- or 12-bit.

↳ The digital data can be right-justified or left-justified.

↳ Major Registers for DAC programming:

- DAC_CR (DAC Control Register)
- DAC_SWTRIGR (DAC Software Trigger Register)
- DAC_DHR (DAC Data Holding Register)
- DAC_DOR (DAC Data Output Register)